Pseudocode & Output-Based Practice

25 Assessment-Style Questions • C/C++/Java/Python-like syntax • Answer Key Included

Instructions: Assume standard integer arithmetic, zero-based array indexing, and normal operator precedence. For questions using ++, assume C/C++/Java-style increment behavior. Try solving before checking the answer key.

1. Increment & assignment

```
int a = 5;
int b = 3;
a = b = a + 2;
print(a, b);
```

2. Post-increment

```
int a = 5;
int b = a++;
int c = ++a;
print(a, b, c);
```

3. Right-to-left assignment

```
int a, b, c;
a = b = c = 10;
b = b + 5;
a = b;
print(a, b, c);
```

4. Operator precedence

```
int x = 2 + 3 * 4;
int y = (2 + 3) * 4;
print(x, y);
```

5. Modulo

```
int x = 17;
int y = 5;
print(x % y);
print(y % x);
```

6. Conditional operator

```
int a = 12;
int b = 7;
int x = (a > b) ? a - b : b - a;
print(x);
```

7. Nested if

```
int x = 15;
if (x > 10) {
if (x % 3 == 0)
print("A");
else
print("B");
} else {
print("C");
}
```

8. For loop

```
int sum = 0;
for (int i = 1; i <= 5; i++)
sum = sum + i;
print(sum);
```

9. Loop with step

```
int x = 1;
for (int i = 1; i <= 5; i = i + 2)
x = x * 2;
print(x);
```

10. Nested loops

```
int count = 0;
for (int i = 1; i <= 3; i++)
for (int j = 1; j <= i; j++)
count++;
print(count);
```

11. Break

```
int sum = 0;
for (int i = 1; i <= 10; i++) {
if (i == 6)
break;
sum = sum + i;
}
print(sum);
```

12. Continue

```
int sum = 0;
for (int i = 1; i <= 6; i++) {
if (i % 2 == 0)
continue;
sum = sum + i;
}
print(sum);
```

13. While loop

```
int n = 20;
int count = 0;
while (n > 1) {
n = n / 2;
count++;
}
print(count, n);
```

14. Array indexing

```
int a[] = {10, 20, 30, 40, 50};
print(a[0] + a[2] + a[4]);
```

15. Array update

```
int a[] = {2, 4, 6, 8};
for (int i = 0; i < 4; i++)
a[i] = a[i] + i;
print(a[1], a[2], a[3]);
```

16. Reverse traversal

```
int a[] = {3, 6, 9, 12};
for (int i = 3; i >= 0; i--)
print(a[i]);
```

17. String indexing

```
string s = "CODE";
print(s[0]);
print(s[2]);
print(length(s));
```

18. String character count

```
string s = "BANANA";
int count = 0;
for (int i = 0; i < length(s); i++) {
if (s[i] == 'A')
count++;
}
print(count);
```

19. Recursion

```
function f(n) {
if (n <= 1) return 1;
return n * f(n - 1);
}
print(f(4));
```

20. Recursive sum

```
function sum(n) {
if (n == 0) return 0;
return n + sum(n - 1);
}
print(sum(5));
```

21. Bitwise AND

```
int a = 6; // 110
int b = 3; // 011
print(a & b);
```

22. Bitwise OR and XOR

```
int a = 5; // 101
int b = 3; // 011
print(a | b);
print(a ^ b);
```

23. Left shift

```
int x = 3;
print(x << 2);
```

24. Right shift

```
int x = 20;
print(x >> 2);
```

25. Mixed expression

```
int a = 4;
int b = 2;
int c = 3;
a = a++ + ++b * c;
print(a, b, c);
```

Answer Key

Q	Answer
1	7 7
2	7 5 7
3	15 15 10
4	14 20
5	2 / 5
6	5
7	A
8	15
9	8
10	6
11	15
12	9
13	4 1
14	90
15	5 8 11
16	12 9 6 3
17	C / D / 4
18	3
19	24
20	15
21	2
22	7 / 6
23	12
24	5
25	13 3 3

Quick exam tip: For output questions, trace variable values line-by-line. Pay special attention to pre/post increment, assignment associativity, loop boundaries, array indices, integer division, and bitwise operators.